Attempt: 2 left (3 attempts every 24 hours)

Sequence Models & Attention Mechanism

Pass • **100**/100 points earned

You must earn 80/100 points to pass.

Re-submit

Question 1:

10/10 points

The network learns where to "pay attention" by learning the values $e^{<t,t'>}$, which are computed using a small neural network:

We can replace $s^{<t-1>}$ with $s^{<t>}$ as an input to this neural network because $s^{<t>}$ is independent of $\alpha^{<t,t'>}$ and $e^{<t,t'>}$.

- [ ] True
- [x] False

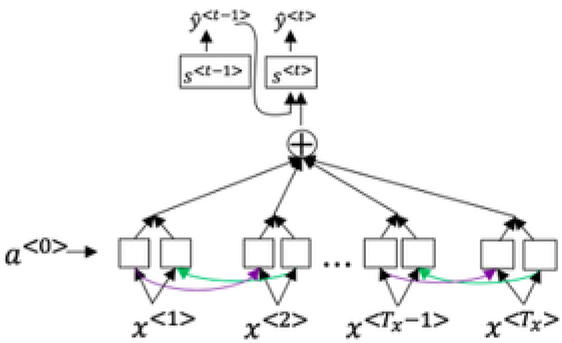
Correct

We can't replace $s^{<t-1>}$ with $s^{<t>}$ as an input to this neural network. This is because $s^{<t>}$ depends on $\alpha^{<t,t'>}$ which in turn depends on and $e^{<t,t'>}$; so at the time we need to evaluate this network, we haven't computed $s^{<t>}$.

Question 2:

10/10 points

Consider the attention model for machine translation.



Further, here is the formula for $\alpha^{<t,t'>}$.

$$\alpha^{<t,t'>} = \frac{\exp(e^{<t,t'>})}{\sum_{t'=1}^{T_x} \exp(e^{<t,t'>})}$$

Which of the following statements about $\alpha^{<t,t'>}$ are true? Check all that apply.

- [ ] $\sum_{t'} \alpha^{<t,t'>} = 0$. (**Note the summation is over t'.**)

- [x] $\sum_{t'} \alpha^{<t,t'>} = 1$. (**Note the summation is over t'.**)

Correct

Correct! We expect $\alpha^{<t,t'>}$ to be larger for activation values that are highly relevant to the value the network should output for $y^{<t>}$.

- [x] $\alpha^{<t,t'>}$ is equal to the amount of attention $y^{<t>}$ should pay to $a^{<t'>}$

Correct

Correct! We expect $\alpha^{<t,t'>}$ to be larger for values of $a^{<t'>}$ that are highly relevant to the value the network should output for $y^{<t>}$.

- [ ] We expect $\alpha^{<t,t'>}$ to be generally larger for values of $a^{<t'>}$ that are highly relevant to the value the network should output for $y^{<t'>}$. (**Note the indices in the superscripts** .)

Question 3: 10/10 points

Suppose you are building a speech recognition system, which uses an RNN model to map from audio clip x to a text transcript y. Your algorithm uses beam search to try to find the value of y that maximizes $P(y \mid x)$.

On a dev set example, given an input audio clip, your algorithm outputs the transcript $\hat{y} =$ "I'm building an A Eye system in Silly con Valley.", whereas a human gives a much superior transcript $y^* =$ "I'm building an AI system in Silicon Valley."

According to your model,

$$P(\hat{y} \mid x) = 1.95 * 10^{-7}$$

$$P(y^* \mid x) = 3.42 * 10^{-9}$$

True/False: Trying a different network architecture could help correct this example.

- (•) True
- ( ) False

Correct

$P(y^* \mid x) < P(\hat{y} \mid x)$ indicates the error should be attributed to the RNN rather than to the search algorithm. If the RNN model is at fault, then a deeper layer of analysis could help to figure out if you should add regularization, get more training data, or try a different network architecture.

Question 4: 10/10 points

True/False: In machine translation, if we carry out beam search using sentence normalization, the algorithm will tend to output overly short translations.

- [ ] True
- [x] False

Correct
In machine translation, if we carry out beam search without using sentence normalization, the algorithm will tend to output overly short translations.

Question 5: 10/10 points

In beam search, if you decrease the beam width B, which of the following would you expect to be true? Select all that apply.

- [x] Beam search will run more quickly.

Correct
As the beam width decreases, beam search runs more quickly, uses up less memory, and converges after fewer steps, but will generally not find the maximum P(y|x).

- [x] Beam search will converge after fewer steps.

Correct
As the beam width decreases, beam search runs more quickly, uses up less memory, and converges after fewer steps, but will generally not find the maximum P(y|x).

- [ ] Beam search will use up more memory.
- [ ] Beam search will generally find better solutions (i.e. do a better job maximizing *P* (*y* | *x*)).

Question 6:

10/10 points

Under the CTC model, identical repeated characters not separated by the "blank" character (_) are collapsed. Under the CTC model, what does the following string collapse to?

kk_eee____ee_p__eeeeeeee_____rrrrr

- ke epe r
- keper
- kkeeeeepeeeeeeeeerrrrr
- keeper

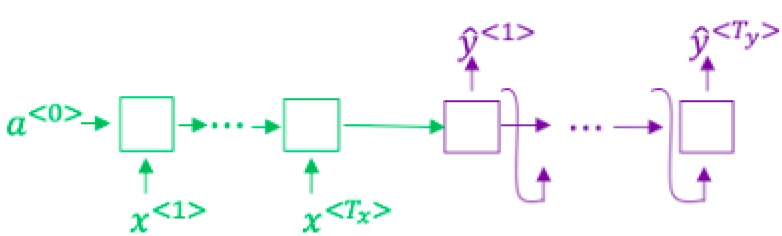
Correct

The basic rule for the CTC cost function is to collapse repeated characters not separated by "blank". If a character is repeated, but separated by a "blank", it is included in the string.

Question 7:

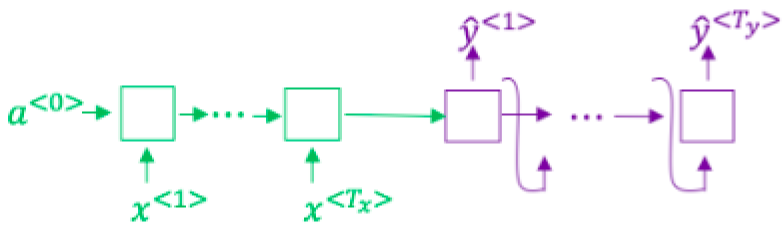
10/10 points

Consider using this encoder-decoder model for machine translation.



Question 7: 10/10 points

Consider using this encoder-decoder model for machine translation.



True/False: This model is a "conditional language model" in the sense that the encoder portion (shown in green) is modeling the probability of the input sentence x.

- True
- False

Correct

The encoder-decoder model for machine translation models the probability of the output sentence y conditioned on the input sentence x. The encoder portion is shown in green, while the decoder portion is shown in purple.

Question 8: 10/10 points

An attention-based model offers the most significant performance improvement over a standard encoder-decoder model under a specific condition. When does the attention model provide the **greatest advantage**?

Question 8:

10/10 points

An attention-based model offers the most significant performance improvement over a standard encoder-decoder model under a specific condition. When does the attention model provide the **greatest advantage**?

- (selected) The input sequence length T_x is large.
- The input sequence length T_x is small.

Correct

This is precisely the scenario where attention models shine. Standard encoder-decoder models struggle with long sequences because they have to compress all information into a single fixed-length vector, creating an "information bottleneck" that causes them to forget the earlier parts of the sequence. The attention mechanism overcomes this by allowing the model to selectively focus on relevant parts of the input, leading to a major performance gain.

Question 9:

10/10 points

Suppose you are building a speech recognition system, which uses an RNN model to map from audio clip x to a text transcript y. Your algorithm uses beam search to try to find the value of y that maximizes $P(y \mid x)$.

Suppose you work on your algorithm for a few more weeks, and now find that for

Question 9:

10/10 points

Suppose you are building a speech recognition system, which uses an RNN model to map from audio clip x to a text transcript y. Your algorithm uses beam search to try to find the value of y that maximizes $P(y \mid x)$.

Suppose you work on your algorithm for a few more weeks, and now find that for the vast majority of examples on which your algorithm makes a mistake, $P(y^* \mid x) > P(\hat{y} \mid x)$. This suggests you should focus your attention on improving the RNN.

- True
- False

Correct

$P(y^* \mid x) > P(\hat{y} \mid x)$ indicates the error should be attributed to the search algorithm rather than to the RNN.

Question 10:

10/10 points

In trigger word detection, $x^{<t>}$ represents the trigger word x being stated for the t -th time

- True
- False

Correct

$x^{<t>}$ represents the features of the audio at time t.